

Euclidian space of n dimensions, $i_1, i_2, \dots, i_n$. These are such that all the ordinary laws of algebra except the commutative law of multiplication apply to the general integral expression q , called a multenion, where

$$q = \sum (x_0 + x_1 i_1 + x_2 i_2 + \dots)(x'_0 + x'_1 i'_1 + x'_2 i'_2 + \dots) \dots, \quad (1)$$

where every x is a scalar. Since any x may be negative we understand this statement to include the ordinary algebraic uses of the signs $+$ and $-$; but we do not understand that any assertion has been made about the use of the sign $\div$ [$q \div r$ can later be understood to be an alternative for qr^{-1}].

LAW II. Scalars are commutative not only with each other but with each of the n primitive unit vectors. [Thus scalars are commutative with multenions.]

LAW III. [i_1^2 means $i_1 i_1$, etc.]

$$-1 = i_1^2 = i_2^2 = \dots = i_n^2.$$

LAW IV. In a product, called a primitive vectorium, $i_a i_b \dots$, of primitive unit vectors, if two adjacent *different* vectors be interchanged, the product merely changes sign.

Such are the assumptions. The rest is development, aided of course by suitable terminology and symbolisation.

DEF. 1. A primitive unit means any one of the following: (1) the scalar 1, (2) any primitive unit vector i_a , (3) any other primitive vectorium $i_a i_b \dots$ [Strictly (3) includes (2) and in a sense (1).]

DEF. 2. An expression of the form

$$p = \sum_{c=1}^n x_c i_c$$

where each x is a scalar, is called a vector, or a V_1 or a $_n V_1$.

DEF. 3. Any multenion is obviously capable of expression in the form $\sum x i_a i_b \dots i_k$ where $i_a, i_b, \dots, i_k$ are m different primitive vectors and m ranges from 0 to n , m being given $V_m q$ means the sum of all parts of the multenion for that value of m . Thus

$$q = V_0 q + V_1 q + V_2 q + \dots + V_n q.$$

$V_a q$ is said to be the part of q of homogeneity a and is called a V_a or a $_n V_a$ or a homogeneous multenion or a hyper vector. [These alternatives are intended to invite suggestions for a settlement of the name. I should have preferred "an a -vector" but this clashes with relativity meanings of 4-vector and 6-vector. In multenions with $n = 4$, there are two kinds of 4-vector $_4 V_1$ and $_4 V_3$.]

$V_0 q$ will generally in words be called the scalar part of q but in the present connection we must not use Sq as an alternative for $V_0 q$. In applications we very decidedly need to be able, in one system of multenions ($n = 4$) to make

no change in the quaternion notation but yet to use both the quaternion notation and the full multenion notation as applying to one symbol q . Thus

$$Sq = V_0q + V_4q, \quad Vq = V_2q.$$

Endless confusion would result from allowing V_0 to be represented by S .

DEF. 4. If $\alpha_1, \alpha_2, \dots, \alpha_a$ are any a -vectors then $V_a\alpha_1\alpha_2\dots\alpha_a$ is called a vectorium. [Hence we have already called such an expression as $i_1i_2i_3$ a primitive vectorium.]

DEF. 5. A linear multenion function of a multenion is called a multenion linity. Similarly for a vector linity or a $_nV_a$ linity, etc. An extended vector linity is a special kind of multenion linity, Φ , defined from a given vector linity, ϕ . Its fundamental property is this: If $\alpha_1, \alpha_2, \dots, \alpha_a$ are any a -vectors whatever, a itself ranging from 0 to n , then

$$\Phi V_a\alpha_1\alpha_2\dots\alpha_a = V_a\phi\alpha_1\phi\alpha_2\dots\phi\alpha_a. \quad (2)$$

[It is not obvious that this remark covers a possibility in general. The definition, which will be modified below, is given here as a hint of the importance of vectoriums and extended linities in our subject.]

Usual Notations.

Scalars (V_0): $a, b, c, g, h, k, l, m, n, t$ ($=$ time), $x, y, z, \mathfrak{J}$ ($= \sqrt{-1}$), θ ($=$ arc), π .

General multenions: p, q, r, s .

Hyper vectors (homog. mult.) and vectoriums:

Homogeneities	V_a	V_b	V_c	V_n	V_2
Hyper vectors	u	v	w	xv	ω
Vectoriums	$\check{\alpha}$	$\check{\epsilon}$	$\check{\eta}$	$x\check{v}$	$\check{\omega}$

(These conventions should be noted and remembered. They immensely simplify the reading of formulae by their property of minimising number of suffixes, etc.)

Vectors (V_1): $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \xi, \eta, \lambda, \mu, \nu, \rho, \sigma, \tau$.

Linear functions: $\xi, \eta, \lambda, \mu, \nu, \varpi, \nu, \phi, \chi, \psi$.

Vectoriums: $\check{\alpha}, \check{\epsilon}, \check{\eta}, \check{l}, \check{\sigma}, \check{\beta}, \check{v}, \check{\omega}$.

Differential symbols: $\delta, \Delta, H, L, d, d_\rho, d_{\check{\rho}b}, d_{s\alpha}, d\alpha, d\check{b}$.

Selective linities, etc.: K, P, Q, R, S, T, U, V .

DEF. 6. Complexes of vectors, hypervectors, multenions, etc., are sufficiently illustrated by multenions. If $q_1, q_2, \dots, q_a$ are a given multenions, then all multenions of the form $\sum_{c=1}^a x_c q_c$ are said to form a complex. If $q_1, q_2, \dots, q_a$

can be made linearly dependent on b and no fewer, then b is called the complexity of the complex. Thus the complexity of the complex of all vectors is n , of all multenions it is 2^n , of all multenions of homogeneity a it is ${}_nC_a$.

DEF. 7. A unit vector or a unit vectorium is one, $\check{a}$, such that $\check{a}^2 = \pm 1$. [The square of every vectorium is a scalar, but u^2 is a scalar in general only in the four cases when $a = 0, 1, n-1$ or n .] Thus all the primitive vectoriums are unit vectoriums.

DEF. 8. Two vectors, α, β , are said to be normal to one another when $V_0\alpha\beta = 0$. Two vector complexes are said to be completely normal to one another when every vector of the one complex is normal to every vector of the other.

Every one of the symbols V_a is a multenion linity. They are all commutative with one another. They are all idempotent, that is $V_a^2 = V_a$, and they all satisfy the equation $V_aV_b = 0$ when $a \neq b$.

There is another system of multenion linities of great utility and simplicity. They are 2^n in number. They are all commutative with each other and with each of the $n+1$ linities V_a . Let P_a be that definite linity of a general multenion, q , which simply reverses the sign of every i_a which occurs in the expression (1) for q . From the well-known properties of linities a product of the P 's such as $P_1P_2P_3$ simply reverses the sign of every i_1, i_2 , and i_3 . These linities are (1) unipotent, that is, if

$$\phi = P_aP_b \cdots P_k, \quad (3)$$

then $\phi^2 = 1$; (2) they are proscriptive, by which is meant that

$$\phi(qr) = \phi q \phi r, \quad (4)$$

where q and r are any two multenions. It is convenient to use P for the product $P_1P_2 \dots P_n$ of all the linities P_a , so that P reverses the sign of every primitive vector.

K is defined, consistently with its quaternion use, in the following manner. Kq is a linity of a general multenion, q , which reverses the sequence of the primitive unit vectors in every primitive vectorium $\check{l}$, of q , and *also* reverses the sign of every primitive unit vector. Thus $K(i_1i_2i_3) = (-i_3)(-i_2)(-i_1) = i_1i_2i_3$. K is (1) unipotent, that is $K^2 = 1$; (2) it is commutative with every V and every P ; (3) it is retrospective, that is $K(qr) = KrKq$. [To prove (3) first prove it when q and r are two primitive vectoriums and then generalise. Such a method of proof is very often applicable and may be referred to as proof by reduction to primitive units.]

$$K(qr) = KrKq. \quad (5)$$

but if ϕ be given by (4), $K\phi(= \phi K)$ is also retrospective. More generally, if $\phi, \psi, \chi, \dots$ are any number of multenion linities, every one of which is either proscriptive or retrospective, then also is the product, $\phi\psi\chi\dots$, proscriptive or retrospective; it is proscriptive when the number of $\phi, \psi, \chi, \dots$ which are retrospective is even, and it is retrospective when that number is odd.

The special retrospective $PK(= KP)$ is of sufficient importance to have a symbol appropriated to it. We shall call it Q . Thus the product of any two of the three P, Q, K is equal to the third.

A caution to users of present methods in relativity and hyperbolic geometry seems desirable. It relates to the discussion of real questions by aid of the imagery

$$\mathfrak{J} = \sqrt{-1}. \quad (6)$$

Let us illustrate by anticipating our mode of treatment of real questions in relativity. We begin with the multenion system based on the primitive generators i_1, i_2, i_3, i_4 . Then we put

$$i_4 = \mathfrak{J}l, \quad (7)$$

and drop out of most of our subsequent work all reference to the original i_4 and $\mathfrak{J}$, using l instead with its property $l^2 = 1$, as opposed to $i_4^2 = -1$. Questions in relativity are then treated by aid of the multiplicative combinations of such expressions as

$$xi_1 + yi_2 + zi_3 + tl. \quad (8)$$

whereby, so long as all the scalars thereby introduced are real, we shall be dealing with real questions in relativity. The general multenion thus constructed contains 16 and not 32 independent real scalars. l is for practical relativity purposes real. We shall call such a multenion system semi-real.

Now return to the meaning above assigned to K .

$$\begin{aligned} K(i_1 i_2 i_3 i_4) &= i_1 i_2 i_3 i_4, \\ \text{and also } K(i_1 i_2 i_3 l) &= i_1 i_2 i_3 l, \\ \text{but whereas } K'(i_1 i_2 i_3 i_4) &= (i_1 i_2 i_3 i_4)^{-1}. \end{aligned} \quad (9)$$

it is not true that the corresponding equation holds when we replace i_4 by l . I have therefore refrained from making the definition of K depend on such equations as (9). It is, nevertheless, a perfectly easy theorem to prove that

$$Kl = l^{-1}. \quad (10)$$

when l is any product of our *original* generators $i_1, i_2, \dots, i_n$. This ceases to be true when l takes the place of i_4 , simply because $\mathfrak{J}^{-1}$ is $-\mathfrak{J}$ and not $+\mathfrak{J}$. It may be noted, however, that there is always a reproscriptive $K\phi = R$ to be found, which is such that for every semi-real vectorium $\check{l}, R\check{l} = \check{l}^{-1}$.

Thus, suppose we replace $i_{a+1}, \dots, i_n$ by $l_{a+1}, \dots, l_n$, where

$$i_c = \mathfrak{J}l_c \quad \text{and therefore} \quad l_c^2 = 1, \quad c = a+1, a+2, \dots, n. \quad (11)$$

Then

$$R = KP_{a+1}P_{a+2} \cdots P_n = QP_1P_2 \cdots P_a, \quad (12)$$

is easily shown to have the required characteristic. We now obviously have

$$P = P_1P_2 \cdots P_n, \quad (13)$$

$$Pu = (-1)^a u, \quad Qu = (-1)^{\frac{1}{2}a(a-1)} u, \quad Ku = (-1)^{\frac{1}{2}a(a+1)} u. \quad (14)$$

Further if we put

$$q_c = V_c q, \quad c = 0, 1, \dots, n, \quad (15)$$

$$Pq = q_0 - q_1 + q_2 - q_3 + \cdots,$$

$$P_{(0)}q = \frac{1}{2}(1+P)q = q_0 + q_2 + q_4 + \cdots,$$

$$P_{(1)}q = \frac{1}{2}(1-P)q = q_1 + q_3 + q_5 + \cdots, \quad (16)$$

$$Qq = q_0 + q_1 - q_2 - q_3 + q_4 + q_5 - \cdots, \quad (17)$$

$$Kq = q_0 - q_1 - q_2 + q_3 + q_4 - q_5 - \cdots. \quad (18)$$

Clearly, we might introduce $Q_{(0)}$, $Q_{(1)}$, and $K_{(0)}$, $K_{(1)}$, corresponding to $P_{(0)}$ and $P_{(1)}$. All the linities recently considered (V_c , ϕ , $K\phi$) may be referred to as the capital linities.

§2. THEOREM OF INDEPENDENCE AND ALLIED ALGEBRAIC FEATURES.—If $q_1q_2 = -q_2q_1$, then q_1 and q_2 are said to be anti-commutative. In the following enunciation, a product of a different multenions from among $q_1, q_2, \dots, q_m$ is said to be odd-formed or even-formed, according as a is odd or even.

If $q_1, q_2, \dots, q_m$ be m multenions such that $q_1^2, q_2^2, \dots, q_m^2$ are all scalars differing from zero, and each pair is anti-commutative; then, if m is even, the 2^m multiplicative combinations are linearly independent; and, if m is odd, they are independent unless the product $q_1q_2 \cdots q_m$ of all of them is a scalar; and, if it is a scalar, 2^{m-1} and only 2^{m-1} of the combinations are independent, and these 2^{m-1} independent combinations may be taken as the odd-formed combinations, or instead as the even-formed combinations, or instead as the combinations from which one assigned multenion, such as q_m , is absent. The theorem is true of linities as well as of multenions.

Thus, if n is even, a general multenion cannot be expressed by fewer than 2^n independent scalars. In the case of $n = 4$, it cannot be expressed by fewer than sixteen independent scalars. In the case of $n = 3$, it can be expressed by four independent scalars, the same number as in the case of $n = 2$. [But to express it thus, we have to impose the condition that $i_1i_2i_3$ is a scalar such as -1 . This is practically Hamilton's procedure in Quaternions.] Either of these cases might be regarded as the case of

QUATERNIONS. I prefer to view Quaternions from another side, namely, as forming the subalgebra of even homogeneities for the case $n = 3$.

In general multienion theory, it is convenient deliberately to prescribe for the case n odd that the 2^n primitive vectoriums shall be independent. This is equivalent to saying that the product $\mathfrak{V} = i_1 i_2 \dots i_n$ of all the primitive vectors is *defined* as independent of the scalar 1, even when n is odd.

Whatever be the value of n , multienions with even homogeneities only form a subalgebra with 2^{n-1} units. When n is odd, this subalgebra can always be identified with the next lower multienion system.

When n is even, the number of independent scalars, 2^n , is a perfect square. In this case the algebra is the equivalent of an algebra of linities with the same number. The characteristic equation, therefore, satisfied by a multienion, may be found. When n is even, the degree of the characteristic equation is $2^{\frac{1}{2}n}$. When n is odd, the degree is $2^{\frac{1}{2}(n+1)}$.

In the case of $n = 4$ or 3 , the characteristic function may be written in a variety of allied forms, of which the two most important are

$$chf_q(x) \equiv (q-x)K(q-x) \cdot (2V_0-1)[(q-x)K(q-x)] \quad (1)$$

$$\equiv (q-x)Q(q-x) \cdot (2V_0-1)[(q-x)Q(q-x)]. \quad (2)$$

The coefficient of every power of x is a scalar which it is not difficult to write down in full. Suppose these scalars are m, m', m'', m''' , thus

$$chf_q(x) = x^4 - m'''x^3 + m''x^2 - m'x + m. \quad (3)$$

This defines the scalars m . We now have the identity

$$q^4 - m'''q^3 + m''q^2 - m'q + m \equiv 0. \quad (4)$$

In the case of the semi-real system suitable to relativity, $chf_q(0)$ is positive or zero, never negative. The positive value of its fourth root is the natural generalisation of the quaternion tensor $+\sqrt{qKq}$. I have been uncertain whether to denote this positive scalar by Tq , thus

$$Tq = +\{qKq \cdot (2V_0-1) \cdot (qKq)\}^{\frac{1}{4}} \quad (1)$$

$$= +\{qQq \cdot (2V_0-1) \cdot (qQq)\}^{\frac{1}{4}}. \quad (2)$$

An objection to doing so is that this is not Hamilton's meaning in the case of his biquaternion $q + \mathfrak{J}r$. His tensor is then imaginary, namely, one of the values of $[(q + \mathfrak{J}r)K(q + \mathfrak{J}r)]^{\frac{1}{2}}$. In our form below, this amounts to saying that the tensor consists of a V_0 part $+aV_4$ part.

It suffices for our purpose to consider that q only has a tensor when qKq is a scalar (V_0). For our semi-real relativity case this must be a real scalar,

and in that case Hamilton's meaning, and the meaning given by (5), coalesce. [If qKq is a scalar, then

$$qKq = (2V_0 - 1) \cdot qKq, \quad (2)$$

and therefore Tq above = $\{(qKq)^2\}^{\frac{1}{4}}$.]

The general value of q for which qKq is a scalar is

$$q = xe^p \quad \text{where} \quad (1 + K)p = 0. \quad (6)$$

The following theorems will now be easily proved by any reader slightly familiar with quaternions.

$$q = \sum i_c V_0 i_c^{-1} q = \sum i_c^{-1} V_0 i_c q, \quad (7)$$

$$\rho = \sum_{c=1}^n i_c V_0 i_c^{-1} \rho = \sum_{c=1}^n i_c^{-1} V_0 i_c \rho, \quad (8)$$

$$V_0(qV_c r) = V_0(rV_c q) = V_0(V_c q V_c r), \quad (9)$$

$$V_0 q r = V_0 r q. \quad (10)$$

whence the cyclic form

$$V_0 q_1 q_2 \dots q_m = V_0 q_m q_1 q_2 \dots q_{m-1},$$

$$V_0(qRr) = V_0(rRq), \quad (11)$$

$$V_0(qKRr) = V_0(rKRq). \quad (12)$$

Here R is any capital retrospective and KR any capital proscriptive. Thus, for the capital linities, not only are they all commutative, but they are all self-conjugate. Conjugacy is defined and tested as in quaternions. Formally, if ϕ is a given multenion linity, its conjugate, ϕ' , is the unique linity for which

$$V_0 q \phi r = V_0 r \phi' q, \quad (13)$$

q and r being arbitrary. When $\phi = \phi'$, ϕ is self-conjugate. More generally, for any ϕ , $\frac{1}{2}(\phi + \phi')$ is the self-conjugate part and $\frac{1}{2}(\phi - \phi')$ is the skew part. More general kinds of conjugate present themselves occasionally. For instance, let R be any one of the capital retroscriptives. Q_R , the R -conjugate of ϕ , may be defined by $V_0 q R \phi r = V_0 r R \phi_R q$. With these meanings of the conjugate, ϕ' , and the R -conjugate, ϕ_R , of ϕ , we have

$$\begin{aligned} (\phi + \psi)' &= \phi' + \psi', & (\phi \psi)' &= \psi' \phi', & (\phi')' &= \phi, \\ (\phi + \psi)_R &= \phi_R + \psi_R, & (\phi \psi)_R &= \psi_R \phi_R, & (\phi_R)_R &= \phi, \\ \phi_R &= R \phi' R, & \phi' &= R \phi_R R. \end{aligned} \quad (14)$$

The general meaning of a conjugate, ϕ_c , of ϕ may be taken to be that it is a unipotent retrospective linity of the linity ϕ . Thus ϕ' is a linity function of the linity ϕ : (1) unipotent because $(\phi')' = \phi$; (2) retrospective because $(\phi \psi)' = \psi' \phi'$; (3) linear because $(\phi + \psi)' = \phi' + \psi'$. I have before

me in MS. a full analysis of the properties of such a conjugate, but must not give it here. [Rq stands to q as ϕ' stands to ϕ .] The capital linities are self- R -conjugate as well as self-conjugate. If $\phi q = pqr$, then

$$\phi'q = rqp, \quad \phi_Rq = Rp \cdot q \cdot Rr, \quad (15)$$

and similarly if $\phi q = p_1qr_1 + p_2qr_2 + \dots$.

Frequently, as in proving the above, we require

$$V_0pr = V_0qr, \quad \text{then} \quad p = q, \quad (16)$$

r being arbitrary, and again

$$V_0\alpha\rho = V_0\beta\rho, \quad \text{then} \quad \alpha = \beta, \quad (17)$$

ρ being arbitrary, and similarly for a general multenion or a vector.

When in a semi-real system R is chosen so that $R\check{i} = \check{i}^{-1}$ for every i , we have that if $q = \sum xi$ and $q' = \sum x'i$, then

$$V_0qRq' = \sum xx',$$

and in particular

$$V_0qRq = \sum x^2.$$

§3. THE COMMUTATION THEOREM.—If $\check{i}_a$ is of homogeneity a , and $\check{i}_b$ of homogeneity b , and if there are exactly c primitive vectors common to the a vectors of $\check{i}_a$ and the b vectors of $\check{i}_b$ it is easy to see that $\check{i}_a\check{i}_b$ is of homogeneity $a + b - 2c$, and that $\check{i}_a\check{i}_b = (-1)^{ab-c}\check{i}_b\check{i}_a$. Hence

$$V_{a+b-2x}uv = (-1)^{ab-x}V_{a+b-2x}vu. \quad (1)$$

[Obviously true when $u = \check{i}_a$, $v = \check{i}_b$ as we see by considering separately the two cases $x = c$, $x \neq c$. It is therefore true for u and v . This is a case of proof by reduction to primitive units.]

(1) may be called a first form of the commutation theorem as it shows us a first effect of commuting u and v in uv . (1) at once leads to what may be called the full form

$$\begin{aligned} uv &= (V_{a+b} + V_{a+b-2} + \dots + V_{a-b})uv \\ &= (-1)^{ab}(V_{a+b} - V_{a+b-2} + \dots + [-1]^b V_{a-b})vu \\ &= (-1)^{a+b}(V_{a-b} - V_{a-b+2} + \dots + [-1]^b V_{a+b})vu. \end{aligned} \quad (2)$$

The following are easy deductions

$$\begin{aligned} \frac{1}{2}[uv + (-1)^{ab}vu] &= (V_{a+b} + V_{a+b-4} + V_{a+b-8} + \dots)uv, \\ \frac{1}{2}[uv - (-1)^{ab}vu] &= (V_{a+b-2} + V_{a+b-6} + \dots)uv. \end{aligned} \quad (3)$$

and these may be written in the reverse order with $(-1)^{ab+b}$ in place of $(-1)^{ab}$. From (3)

$$\begin{aligned} \frac{1}{2}(uv + vu) &= (V_{a+2} + V_{a-2})uv = (V_{a+2} + V_{a-2})vu, \\ \frac{1}{2}(uv - vu) &= V_a \cdot uv = -V_a \cdot vu. \end{aligned} \quad (4)$$

which has important applications to infinitesimal rotations. Other special cases are

$$u^2 = (V_0 + V_4 + V_8 + \dots)u^2, \quad (5)$$

$$\omega^2 = (V_0 + V_4)\omega^2, \quad (6)$$

$$\alpha\beta = (V_0 + V_2)\alpha\beta = (V_0 - V_2)\beta\alpha, \quad (7)$$

$$V_2\alpha\beta = -V_2\beta\alpha, \quad V_0\alpha\beta = V_0\beta\alpha, \quad (8)$$

$$V_2\alpha\beta = \frac{1}{2}(\alpha\beta - \beta\alpha), \quad V_0\alpha\beta = \frac{1}{2}(\alpha\beta + \beta\alpha). \quad (9)$$

§4. ROTATIONS OF THREE KINDS.—Suppose we find n multenions $p_1, p_2 \dots p_n$ such that for all purposes of Laws I to IV they may take the place of $i_1, i_2 \dots i_n$, are we to call the substitution of the multenions p for the vectors i a rotation? In Euclidean space of three dimensions we should not always so affirm. If i, j, k should represent, as sometimes in quaternions they do, three directions in a rigid body we could not move the body so that i became $-i$ and j and k remained unaltered; but they would continue to satisfy the conditions of the laws. We should call the transformation a perversion or reflection rather than a rotation. As the algebra of multenions is (n even) a full linity algebra, or (n odd) a sub-algebra of such a linity algebra, we will assume that given q , q^{-1} can be obtained uniquely in general so that $qq^{-1} = 1 = q^{-1}q$ (though in certain singular cases we have to say instead that q_0 can be found so that $qq_0 = 0 = q_0q$). In the general case we will say that q is vertible (*i.e.*, it has an inverse q^{-1}); in the singular case that q is non-vertible.

Theorem concerning multenion rotation. (1) When q is a given vertible multenion, then in the four fundamental laws the primitive vectors $i_1, i_2, \dots i_n$ may be replaced respectively by

$$p_1 = qi_1q^{-1}, \dots, \quad p_n = qi_nq^{-1}, \quad (1)$$

and the effect of the replacement is to change any multenion r to

$$r' = qrq^{-1}. \quad (2)$$

(2) Conversely if $p_1, p_2 \dots p_n$ are n multenions such that

$$-1 = p_1^2 = p_2^2 = \dots = p_n^2, \quad (3)$$

and every pair is anti-commutative, that is

$$p_1p_2 + p_2p_1 = 0, \text{ etc.}, \quad (4)$$

and when n is odd

$$p_1p_2 \dots p_n = i_1i_2 \dots i_n, \quad (5)$$

then there exists a vertible multenion, q , for which (1) and (2) are true; r' now meaning the multenion obtained from the given multenion r by replacing $i_1, i_2, \dots i_n$ by $p_1, p_2, \dots p_n$ respectively.

This kind of rotation is scarcely what we seek. For instance, it does not in general leave V_c unchanged, that is, it is not in general true that

$$qV_c r \cdot q^{-1} = V_c(qr q^{-1}). \quad (6)$$

Multenion rotation qrq^{-1} is operation by a multenion linity $\phi = q()q^{-1}$ such that $\phi'\phi = 1 = \phi\phi'$, and conversely.

Next, imposing the condition that (6) is to be satisfied, we find that q must be the product of any number of vectors, and that if it is such a product (6) is satisfied. We may next find what is the effect of infinitesimally changing the vectors and attempt to build up the finite continuous group of which that infinitesimal transformation is the element. The infinitesimal rotation is the transformation effected by changing q to $q + \frac{1}{2}(\omega q - q\omega)$, where ω is infinitesimal and the corresponding finite rotation is $e^\omega()e^{-\omega}$, where ω is finite. ω is a perfectly arbitrary ${}_nV_2$, infinitesimal of course for the first case. When ω is not singular ($\omega^2 = 0$) its general form may be put

$$\omega = a\varepsilon_1\varepsilon_2 + b\varepsilon_3\varepsilon_4 + \dots,$$

where $\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4 \dots$ are orthogonal unit vectors,

$$e^\omega \equiv (\cos a + \varepsilon_1\varepsilon_2 \sin a)(\cos b + \varepsilon_3\varepsilon_4 \sin b) \dots$$

and the factors here are commutative. Putting the last in the form

$$e^\omega = \varepsilon_1(\varepsilon_1^{-1} \cos a + \varepsilon_2 \sin a)\varepsilon_3(\varepsilon_3^{-1} \cos b + \varepsilon_4 \sin b) \dots,$$

we see that e^ω is the product of an *even* number of vectors. The general form of ω when it is singular is not simple, but I have a complete solution before me in MS.

To show that the infinitesimal rotation is as stated will not occupy too much space. Suppose

$$r' = \alpha\beta \dots \lambda \cdot r \cdot \lambda^{-1} \dots \beta^{-1}\alpha^{-1} = prp^{-1},$$

where the vectors $\alpha, \beta, \gamma, \dots$ are assumed to be unit vectors, as that does not affect the transformation. Let $\alpha, \beta, \dots$ change to $\alpha + d\alpha, \beta + d\beta, \dots$, still remaining unit vectors.

$$dr' = d(prp^{-1}) = dpp^{-1} \cdot r' - r'dpp^{-1}.$$

That dpp^{-1} is a ${}_nV_2$ is straightforwardly proved thus

$$dpp^{-1} = d\alpha\alpha^{-1} + \alpha \cdot d\beta\beta^{-1} \cdot \alpha^{-1} + \alpha\beta \cdot d\gamma\gamma^{-1} \cdot \beta^{-1}\alpha^{-1} + \dots.$$

Every term here is a ${}_nV_2$; for example, consider the third term. Since γ and $\gamma + d\gamma$ are unit vectors, $V_0 d\gamma\gamma^{-1} = 0$, and therefore $d\gamma\gamma^{-1}$ is a ${}_nV_2$. It follows that the third term is a ${}_nV_2$. If we put u, u' for r, r' and $\frac{1}{2}d\omega$ for dpp^{-1} ,

$$du' = \frac{1}{2}(d\omega u' - u'd\omega) = V_0 d\omega u'.$$

This is the very simple, very useful form in which true infinitesimal rotation

presents itself in our present subject. A rotation of vectors ρ is a linity $\phi\rho$ of ρ such that $\phi'\phi = 1$.

When without qualification we speak of rotation we shall understand it to mean $e^\omega()e^{-\omega}$ (a continuous group).

The skew part of any vector linity, ϕ , is of the form of an infinitesimal rotation. Thus, introduce the ζ notation of quaternions,¹ by which

$$F(\zeta, \zeta) = \sum_{c=1}^n F(i_c, i_c), \quad (7)$$

where F is bilinear in its two members. We have

$$\rho = -\zeta V_0 \zeta \rho, \quad (8)$$

$$\phi' \rho = -\zeta V_0 \zeta \phi' \rho = -\zeta V_0 \rho \phi \zeta. \quad (9)$$

The skew part is

$$\frac{1}{2}(\phi - \phi')\rho = \frac{1}{2}V_1(V_2\zeta\phi\zeta)\rho = V_1\omega\rho. \quad (10)$$

Here we have assumed that

$$V_1(V_2\alpha\beta)\gamma = \alpha V_0\beta\gamma - \beta V_0\gamma\alpha,$$

much as in quaternions. This is not difficult to prove, and we shall in §7 enunciate much more general theorems of which this is a particular case. The pure part does not similarly reduce. We have

$$\frac{1}{2}(\phi + \phi')\rho = -\frac{1}{2}(\zeta V_0 \rho \phi \zeta + \phi \zeta V_0 \rho \zeta). \quad (11)$$

§ 5. VECTORIUMS. Let

$$\alpha^{(a)} = \alpha_1 \alpha_2 \dots \alpha_a, \quad \beta^{(b)} = \beta_1 \beta_2 \dots \beta_b, \quad \gamma^{(c)} = \gamma_1 \gamma_2 \dots \gamma_c.$$

Then in the expression $V_{a+b+c}(\alpha^{(a)}\beta^{(b)}\gamma^{(c)})$ it is obvious by reduction to primitive units that within the brackets we may at will write either $\alpha^{(a)}$ or $V_a\alpha^{(a)}$, and $\beta^{(b)}$ or $V_b\beta^{(b)}$, and $\gamma^{(c)}$ or $V_c\gamma^{(c)}$; the value of the expression is unaltered by such insertion or removal of V_a , V_b , or V_c . The statement, of course, is in general only true of this particular part, V_{a+b+c} of the product of highest homogeneity. Next, it is similarly evident that if we change, say, α_2 to $x\alpha_2 + y\alpha'_2$ we simply alter the corresponding vectorium similarly, that is

$$V_a\alpha_1(x\alpha_2 + y\alpha'_2)\alpha_3 \dots \alpha_a = xV_a\alpha_1\alpha_2\alpha_3 \dots \alpha_a + yV_a\alpha_1\alpha'_2\alpha_3 \dots \alpha_a.$$

Again, if in the vectorium we interchange two consecutive members such as α_2, α_3 we merely change the sign of the whole. For we may successively write instead of $\alpha_2\alpha_3$

$$\alpha_2\alpha_3, \quad V_2\alpha_2\alpha_3, \quad -V_2\alpha_3\alpha_2, \quad -\alpha_3\alpha_2.$$

Finally, by a familiar simple process we may interchange any two members by successive interchanges of neighbours, and as a final result, again, the vectorium changes in sign merely. Thus $V_a\alpha^{(a)}$ is “combinatorial.” The

¹See *Utility of Quaternions*.

common simple properties of a combinatorial multilinear function are probably well known to my readers and may be assumed. [Grassmann apparently first, in a perfectly general manner, appreciated the value of, and fully established these properties. Indeed, $V_a \alpha^{(a)}$ is both algebraically and geometrically identical in meaning with Grassmann's combinatorial "product." I should like for a second time to express my unbounded admiration for Grassmann's great pioneer work, though I cannot help believing that the actual form of the matters he dealt with has been put out of court by later developments.]

We may note the following theorem in passing. It is our form of one of Grassmann's products. If $\check{p}$ is a vectorium of homogeneity g , then

$$V_{a+b+c}(V_{g+a+b}\check{p}uv)(V_{g+c}w\check{p}) = V_{a+b+c}(V_{g+a}\check{p}u)(V_{g+b+c}v\check{p}w).$$

Of u, v, w the middle one, v , may be passed over from the first to the second bracket-pair.

The following is wanted immediately and is constantly in request. It may be thought of as a passage into or out of the operation of a $V_a()$ [whether as pre-factor or post-factor] of either $\check{v}$ or p_n . We have

$$\check{v}V_a q = V_{n-a}\check{v}q, \quad V_a q \cdot \check{v} = V_{n-a}(q\check{v}).$$

Also, if p_n is any multienion of homogeneity n , $p_n = x\check{v}$, and, therefore, in the present process, p_n behaves exactly as does $\check{v}$, that is

$$p_n V_a q = V_{n-a} p_n q, \quad V_a q \cdot p_n = V_{n-a} q p_n.$$

Suppose there are n given linearly independent vectors $\alpha_1, \alpha_2, \dots, \alpha_n$. Let $\alpha^{(a)}$ be the product of any a of them, and $\alpha^{(n-a)}$ the product of the rest in such sequence that $V_n \alpha^{(a)} \alpha^{(n-a)}$ always retains the same value. Let

$$\begin{aligned} \check{\alpha} &= V_a \alpha^{(a)}, \\ \check{\alpha} &= V_{n-a} \alpha^{(n-a)} \cdot V_n^{-1} \alpha^{(a)} \alpha^{(n-a)}. \end{aligned} \tag{1}$$

[By a convenient, strictly speaking indefensible, custom, since V_n^{-1} really = ∞ , we mean by $V_n^{-1} \alpha^{(a)} \alpha^{(n-a)}$ the reciprocal $(V_n \alpha^{(a)} \alpha^{(n-a)})^{-1}$.]

There are 2^n linearly independent [proof of linear independence here omitted] values of $\check{\alpha}$, and they will be called the 2^n vectoriums formed from $\alpha_1, \alpha_2, \dots, \alpha_n$. [The term may, of course, be used when $\alpha_1, \alpha_2, \dots, \alpha_n$ are not linearly independent.] The 2^n corresponding values of $\check{\alpha}$, which are also linearly independent, will be called the normal reciprocals of the vectoriums, $\check{\alpha}$. They are so called because

$$V_0 \check{\alpha} \check{\alpha} = 1, \quad V_0 \check{\alpha}_1 \check{\alpha} = 0,$$

$\check{\alpha}_1$ being a second (not corresponding to $\check{\alpha}$) $\check{\alpha}$. [Thus, when $n = 4$, there are

sixteen equations of the type $V_0 \check{\alpha} \hat{\alpha} = 1$, and 240 equations of the type $V_0 \check{\alpha}_1 \hat{\alpha} = 0$.] Both equations are proved quite simply from (1) by passing $p_n = V_n^{-1} \alpha^{(a)} \alpha^{(n-a)}$ out of the operation of V_0 in $V_0 \check{\alpha} \hat{\alpha}$.

Let now $q = \sum x \check{\alpha} = \sum y \hat{\alpha}$. From the normal reciprocal relations we at once get $x = V_0 q \hat{\alpha}$, $y = V_0 q \check{\alpha}$. Hence

$$\begin{aligned} q &= \sum \hat{\alpha} V_0 q \hat{\alpha} = \sum V_a \alpha^{(a)} \cdot V_n q \alpha^{(n-a)} \cdot V_n^{-1} \alpha^{(a)} \alpha^{(n-a)}, \\ &= \sum \hat{\alpha} V_0 q \check{\alpha} = \sum V_0 (q V_a \alpha^{(a)}) \cdot V_{n-a} \alpha^{(n-a)} \cdot V_n^{-1} \alpha^{(a)} \alpha^{(n-a)}. \end{aligned} \quad (2)$$

the last result in the first line coming from a new application of the p_n theorem. If in (2) we put $q = u$, getting a more particular result, we should at once see that (2) is analogous to the quaternion theorems

$$\rho S \alpha \beta \gamma \equiv \alpha S \beta \gamma \rho + \dots \equiv V \beta \gamma S \alpha \rho + \dots$$

A particular case ($a = 1$) of (1) is

$$\begin{aligned} \hat{\alpha}_1 &= V_{n-1} \alpha_2 \alpha_3 \dots \alpha_n \cdot V_n^{-1} \alpha_1 \alpha_2 \dots \alpha_3 \\ &= -V_{n-1} \alpha_1 \alpha_3 \dots \alpha_n \cdot V_n^{-1} \alpha_1 \alpha_2 \dots \alpha_3 = \text{etc.} \end{aligned} \quad (3)$$

This is clearly the solution of the problem. Given

$$S \alpha_c \hat{\alpha}_c = 1, \quad c = 1, 2, \dots, n, \quad S \alpha_b \hat{\alpha}_c = 0, \quad b \neq c; \quad b, c = 1, 2, \dots, n,$$

what are the vectors $\hat{\alpha}_c$ in terms of the vectors α_c ? And, of course, we have a precisely similar solution for α_c in terms of $\hat{\alpha}_c$. It appears, then, that the relations between the quantities $\check{\alpha}$ and the quantities $\hat{\alpha}$ are symmetrical. No doubt we have here established this if only it be evident what is the precise significance of the word "symmetrical" in this connection. The need of the caution may be illustrated thus. If we put $\hat{\alpha} = V_a \hat{\alpha}_1 \hat{\alpha}_2 \dots \hat{\alpha}_n$, we are tempted to say that $\hat{\alpha}$ must be $V_a \hat{\alpha}_1 \hat{\alpha}_2 \dots \hat{\alpha}_a$, but we shall find that we do not thereby always obtain $S \hat{\alpha} \hat{\alpha} = 1$. The true value of $\hat{\alpha}$ is

$$\begin{aligned} \hat{\alpha} &= V_a \hat{\alpha}_a \hat{\alpha}_{a-1} \dots \hat{\alpha}_1 = Q V_a \hat{\alpha}_1 \hat{\alpha}_2 \dots \hat{\alpha}_a, \\ \text{when} \quad \hat{\alpha} &= V_a \alpha_1 \alpha_2 \dots \alpha_a. \end{aligned} \quad (4)$$

If the α_c are taken as the i_c , it is obvious that the $\hat{\alpha}$ are the primitive vectoriums, and, from the normal reciprocal relations, it is then obvious that the $\check{\alpha}$ are also the primitive vectoriums (with different signs in accordance with) $\hat{\alpha} = K \check{\alpha}$.

[Thus $0' = 1$, $1' = -1$, $2' = -1/2!$, $3' = +1/3!$ etc., the signs running $+- - + + - - + + - - \dots$]

$$\sum F(i_a, i_a^{-1}) = a' F(V_a \zeta^{(a)}, V_a \zeta^{(a)}), \quad (6)$$

$$u = \sum i_a V_0 i_a^{-1} = a' V_a \zeta^{(a)} \cdot V_0 i_a \zeta^{(a)}. \quad (7)$$

In (6) put the second $V_a \zeta^{(a)} = \sum \hat{\alpha} V_0 \hat{\alpha} \zeta^{(a)}$. Thus,

$$a' F(V_a \zeta^{(a)}, V_a \zeta^{(a)}) = a' \sum (V_a \zeta^{(a)} V_0 \hat{\alpha} \zeta^{(a)}, \hat{\alpha}) = \sum F(\hat{\alpha}, \hat{\alpha}).$$

We might have modified the first $V_a \zeta^{(a)}$ instead of the second. Thus collecting results

$$\begin{aligned} a' F(V_a \zeta^{(a)}, V_a \zeta^{(a)}) &= \sum F(i_a, i_a^{-1}) \\ &= \sum F(\hat{\alpha}, \hat{\alpha}) = \sum F(\alpha, \hat{\alpha}). \end{aligned} \quad (8)$$

($n C_a$ terms in each summation.) Just as in quaternions $F(\zeta, \phi \zeta) = F(\phi' \zeta, \zeta)$ so now if ϕ is a $n V_a$ linity

$$a' F(V_a \zeta^{(a)}, \phi V_a \zeta^{(a)}) = a' F(\phi' V_a \zeta^{(a)}, V_a \zeta^{(a)}). \quad (9)$$

If ϕ is a multenion linity

$$\sum_{a=0}^n a' F(V_a \zeta^{(a)}, \phi V_a \zeta^{(a)}) = \sum_{a=0}^n a' F(\phi' V_a \zeta^{(a)}, V_a \zeta^{(a)}). \quad (10)$$

For $a = 0$ we must put $a' = 1$, and for each $V_a \zeta^{(a)}$ we must also put 1 [see (7) above].

§ 6. EXTENDED VECTOR LINITIES.—The theorems of §§ 6, 7 find continuous applications in the multenion treatment of differential invariants. Such treatment may be considered to begin in §8, and is explicitly introduced in §9.

Let ϕ be a given vector linity. The extended linity (a multenion linity) which may for most purposes, but not for all, also be denoted by ϕ , is here defined thus. Let $i_a = i_g i_h \dots i_l$ be any primitive unit of homogeneity a , a itself being arbitrary. Then

$$\phi V_a i_a = V_a \phi i_g \phi i_h \dots \phi i_l.$$

Now change the n primitive vectors $i_1, i_2, \dots, i_n$ to any other n vectors by such steps as replacing i_1 by $x i_1 + y i_2$ (or, if i_1, i_2 have already changed to $\varepsilon_1, \varepsilon_2$, the change should be from ε_1 to $x \varepsilon_1 + y \varepsilon_2$). Such a change simply multiplies each side of the defining equation (when i_1 occurs in it) by x ; the meaning is therefore unaltered. It follows that $\phi V_a \alpha^{(a)} = V_a (\phi \alpha)^{(a)}$ where a is arbitrary and the vectors α are arbitrary. If, then, $(\phi), (\psi)$ are the extended linities of ϕ, ψ , we see that $(\phi)(\psi)$ is the extended linity of $\phi \psi$, for

$$(\phi)(\psi) V_a i_g i_h \dots i_l = (\phi) V_a \psi i_g \psi i_h \dots \psi i_l = V_a \phi \psi i_g \phi \psi i_h \dots \phi \psi i_l.$$

From this again it follows that the extended linity of ϕ^{-1} is $(\phi)^{-1}$.

Let us use either (ϕ) or ϕ for the extended linity of ϕ for the present. What we have just proved may be put

$$(\phi)(\psi) = (\phi\psi), \quad (\phi)^{-1} = (\phi^{-1}).$$

We are now about to prove a similar proposition concerning ϕ' the conjugate, namely

$$(\phi)' = (\phi').$$

By (7) of §5 we have

$$\phi u = a' \phi V_a \zeta^{(a)} V_0 u \zeta^{(a)}, \quad (\phi)u = a' V_a (\phi \zeta)^{(a)} V_0 u \zeta^{(a)}. \quad (1)$$

Let u, u_0 be two arbitrary multenions of homogeneity a . Operate on (1) by $V_0 u_0()$

$$V_0 u_0 (\phi)u = a' V_0 u_0 (\phi \zeta)^{(a)} V_0 u \zeta^{(a)}.$$

Modifying the left by the fundamental rule of the meaning of $(\phi)'$, and modifying the right by the rule $F(\zeta, \phi \zeta) = F(\phi' \zeta, \zeta)$ applicable to a vector linity. We obtain

$$V_0 u (\phi)' u_0 = a' V_0 u_0 \zeta^{(a)} V_0 u (\phi' \zeta)^{(a)}.$$

Since u is arbitrary, by (16) of §3

$$(\phi)' u_0 = a' V_a (\phi' \zeta)^{(a)} V_0 u_0 \zeta^{(a)},$$

which by comparison with (1) proves the assertion that $(\phi)' = (\phi')$.

The following theorems are also true

$$\phi V_{a+b} uv = V_{a+b} \phi u \phi v, \quad (2)$$

$$\phi V_{a-b} uv = V_{a-b} \phi u \phi'^{-1} v, \quad (3)$$

when ϕ is non-vertible we must change v to $\phi' v$. An alternative, less convenient than (3) generally, is thus

$$\phi V_{a-b} u \phi' v = V_{a-b} \phi u v, \quad (4)$$

but this alternative is true without qualification. $\check{v}$ having its usual meaning $(i_1, i_2, \dots, i_n)$ it is easy to see that $\phi \check{v} = m \check{v}$ where m is the determinant of ϕ . Putting $u = \check{v}$ in (3) and (4) we obtain

$$\begin{aligned} \check{v}^{-1} \phi (\check{v} v) &= m \phi'^{-1} v, \\ \text{or} \quad \check{v}^{-1} \phi (\check{v} \phi' v) &= m v. \end{aligned} \quad (5)$$

[Added February, 1921.—Closely connected with the extension in §6 from vector operation to multenion operation, of a given vector linity, is a second kind of such extension. If ϕ_E is the extended (or, as I now prefer to say